

Alex Kiefer: “Resilience from an active inference perspective”

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Duration: 25:23 | Uploaded: Unknown Transcript method: auto_caption

all right keeping with the co-authorship theme next talk is a 25 minute recorded talk by Alex kefir resilience and active inference 2024 this is Alex kefir this is Alex kefir I'm here to present some ideas on on resilience and active inference um so this is largely just a unpacking in a careful way of some Concepts from this paper from 2022 that I was involved in um but also will explore some some novel directions um that this uh this thinking has taken since then so the concept of resilience really if you look at the emology it's just it means like the capacity to recoil essentially or to or to bounce back or just to survive through hardship and change um so obviously this concept is sort of a fundamental one if you're looking at uh life life forms um dynamical systems more generally um in the context of like um ecosystems and sustainability it's an important concept um so that's why we're interested in this um it's actually also sort of a fundamental Concept in Material Science so so since this paper has come out I've been approached a couple of times by by journals of Material Science for um for follow-ups and such um so as I said the the fundamental concept has to do with recoiling or bouncing back um but there's sort of there are some peripheral sort of different facets of this of this meaning and we explored all of these in this paper um so roughly speaking you can think of resilience in sort of three ways um so first there's inertia um so inertia I mean I guess literally just means really uh uh sort of not changing um so persistence uh but you can also think of it as resistance in the sense that right unless you're unless you're a exist in a vacuum uh then you're a system that's embedded in some broader system and inertia is basically continuing on your way despite changes uh that might that might occur or interactions with that broader system um so if you think of the red arrow as representing the path of a of a subsystem or an H through its state space then and The Black Arrow is some kind of potential disturbance um then the top diagram here illustrates inertia um the central meaning again of resilience arguably is elasticity or the ability to bounce back so in that case you undergo some perturbation or change um and you end up recovering your initial doesn't have to be that you recover your exact initial

position but the point is you make your way back to a reasonable state in terms of the kind of system that you are um then there's also this sort of third thing called we call plasticity or adaptation um so in this case you begin in one configuration uh the red arrow at the beginning here and after the perturb uh the perturbation or disturbance you end up in a slightly different configuration so that should be read not as just that you change your state like you update your um your beliefs in the time scale of inferences or something but you actually you actually wind up somewhat adjusting your model in response to this change um so this is more like something like Allostasis um whereas plasticity is returning to the same state um so i' say there are some close relationships just to get into some philosophical conceptual analysis for a second here um the first thing uh inertia probably doesn't really exist in its pure form right in nature in a way or it if it does it reduces to sort of elasticity so you know I mean presumably if you if you perturb a system um then you have to cause some change in the system however slight um otherwise there hasn't been a physical interaction right uh and inertia is going to look in practice just like the capacity to sort of instantly recover um from that tiny from that tiny change um there there is in the limit case there is sort of of different meaning here because again there might just not be a change but I think um I think in the context of realistic systems there will always be some some interaction um but arguably these two these two concepts are very close um plasticity seems like um on the face of it like a different thing because um well first of all uh the ability to be to be plastic in this sense sort of explains how you could come to have this property of elasticity um so if you look at the example of neuroplasticity for example um it's the ability to adapt to change and to basically update your your synaptic weights to arrive at a new model that allows you to be uh in part allows you to be resilient in these first two senses um but there's another sense in which if you just assume that there's some property of the system that doesn't change through this um through this process which I think is a reasonable assumption in the case of uh you know biological systems um then there's going to be some kernel of the system that survives through change and then you're back to your sort of original definition um okay so if we look at this concept through the lens of the free energy principle um I mean first of all it turns out to be a very fundamental at you know as is often the case when you try to unpack um sort of uh fundamental properties of of systems in nature in terms of the FP and active inference uh you find that there's a fairly direct mapping from the properties that the FP attributes to systems or um the types of systems that are well described by the FP and this uh and this concept of resilience um so I mean Carl Friston has said in discussions of this that essentially uh systems well described by the FDP are resilient um uh full

stop so uh but we can look at at the three uh facets of the meaning of resilience here so um certainly systems described by the Fe are they have this property of inertia in the sense that um by definition right they're things that have managed to persist so they limit their exposure to uh to unlikely States unlikely observations um but they're also sort of by definition elastic systems um because because they have this attracting set so um having an attracting set of States means that um you know of course there might be changes or perturbations that are sort of catastrophic that destroy the system but as long as the system is not destroyed then by definition it's going to make it make its way back to the subtracting set of States um so resilience in the core sense is really a um almost a defining feature of systems that are described by the FP uh and then finally plasticity you can think of as something like basian model selection um on some time scale and so that that might seem like something extra but it's only extra if the multiscale aspect of the FP is extra and I would argue that that it really isn't because at its heart the FP is a scale-free description of things and this multiscale and scale free are sort of uh sort of um connected Notions um so that's plasticity um okay but so most of the interest I'd say of this lies in the the mapping of resilience onto this this formalism um of active inference so uh a key concept uh in this analysis is the notion of degeneracy and this is is a uh somewhat technical term um here we're follow we mainly follow on the definition of and and in fact the analysis of degeneracy in terms of active inference in uh Sajid at all 2020 this is a great paper on this topic um so here we have just the expression for variational free energy um in the top row um is a expression in terms of basically a Helm hols free energy so it's an energy term minus an entropy term uh uh so if you just look at that for a second um the we've identified uh again following this paper from 2020 we've identified degeneracy with the entropy term um so we'll get into why that is in a second um but basically uh if the concept of degeneracy is useful redundancy um so this you can think of this as like complexity that that earns its keep or that's worth having um so complexity in this this bottom breakdown of variational free energy uh you uh you analyze it in terms of complexity which is a the the difference the K Divergence between the prior belief uh Over States the the generative density Over States and the approximate posterior so this is kind of like how how difficult is it or how much change has to uh does the system have to undergo in order to represent the states that are um inferred from some observations um what's what's what's the departure between that posterior distribution and the prior dist distribution Over States uh that's complexity um minus accuracy um the the the expected uh log probability of observations given States um so what's the connection between complexity and this idea of

useful redundancy uh sorry of just redundancy um there's there's a assumption here that complexity and redundancy are closely related um I think the basic idea is that so redundancy uh in this context is quite literally just duplication of States um so the I'd say the classic example is like an error correcting code so if you're trying to transmit some uh a bit of information over a channel uh you might you might send 10 copies of a bit in case in case some of those bits get flipped or corrupted uh and you want to make sure that the that the message gets transmitted um so that's redundancy it increases the complexity because um uh well one way of looking at it is that it uh it it increases the description length uh right this this K Divergence is like how how how much how much more do you have to say in order to communicate that an event drawn from this Q distribution has occurred um G given that you assume this P distribution um so uh so you can think of it as well what these these extra bits are are the Redundant bits um so there there's a sort of conceptual connection between complexity and redundancy um and then degeneracy on this first definition is just the complexity that serves some purpose and we'll get into that in the moment um it's just also important to note that there's this definition of um degeneracy as partial redundancy in the in the literature uh and this is partial overlap and function so um the idea is that um redundancy exists in a system uh sorry degeneracy resists uh exists to the extent that multiple states in the system serve some function in common but then those States also have um otherwise uh have distinct or disjoint sets of of functions um and in practice I think these two almost always line up so there might be sort of uh limit edge cases in which um in which they're different but in but in practice um right if you have multiple States they'll always have some different properties and therefore um perform some different functions okay so so this is one of the core things that we pointed out in this paper um of ours so um this this actually sorry this breakdown is from the sjad all 2020 paper uh but we sort of have a different gloss on the Central Center term here um so this is the complexity term in the variational free energy you can break this down in terms of um further break this down in terms of this uh expectation under Q of the negative log probability of States under the generative model um so this is called a cost term in the 2020 paper um we describe this in epistemic terms as as an uncertainty term so this is uh this is basically uh you can you can read this as the Shannon uh minimum description length of the of the states given the generative density um but it's sort of the expected surprise um how surprising are are these states that I that I infer uh using the variational posterior under my generative model um so that's an uncertainty term uh and then we subtract this degeneracy or the entropy over um over the variation of posterior Over States

um so I mean first of all sorry I realized that I didn't quite make the the link here between um degeneracy and entropy so why is this the case uh why do we identify these uh it's it's because um this means if if the if if this Q distribution Over States is higher in entropy it means that there's a um degenerate mapping in the likelihood from states to observations right so meaning for multiple States uh you can get the same observation um and so then when you invert that likelihood uh mapping in order to recover this posterior um this means that for one observation you might have several um several different states and the spread of the of the distribution Over States of course is just the entropy so that's that's the link to to generacy there uh so so this complexity term is uncertainty minus degeneracy um one reason that's interesting is that when this complexity is zero um it's not as if the system is completely certain of what's going on um the uncertainty term actually remains it's just balanced by this degeneracy in other words all of the uncertainty in this ideal case um is is useful um and so I think this really this really captures what what degeneracy in this context amounts to um it's uncertainty that's useful and is it's useful precisely because there's some amount of uncertainty that's inherent in the source of the information you're trying to model right so um here p is the the in this context is the generative model that the system has of um it's the sort of internal generative model of of the external States um and so there will be some right there'll be some entropy some Source entropy associated with that system that that you're trying to model and um so long as you match your variational posterior matches that degree of uncertainty um then you're in sort of an optimal um an optimal model so in other words for maximum model goodness try to ensure that um P equals Q um so uh this ties nicely into the idea plasticity resilience is plasticity so your goal here is to adjust the posterior to match the generative model essentially um so you can do that um you know in in like perceptual inference uh you can try to infer a state uh a q distribution this is again an approximation to the posterior the true posterior over over those States under the generative model um but equally you can adjust the generative model in in a in a multiscale setting you can learn learn or adjust update the parameters of the generative density in order to match what's in the external World um so right this this complexity term um is affected by changes to p as well as changes in Q and so I think the fundamental idea that I had here was that um in an ideal uh sort of optimal system under the free energy principle you should expect um first of all P will be adjusted so as to model the environment for sort of um basic you know um free energy principle SLG good regulator theorem types of reasons uh and then Q will also be adjusted to match to match P um so this involves um this cybernetic principle called The Law of requisite variety which is sort of a um

um it's closely related to the good regul good regulator theorem um but basically it just says that you know if you're trying to regulate or control some States then you need uh at least as much uh variety or uh in probabilistic terms that that translates to variance um or or entropy um you need as much Variety in the controller as you have in the in the environment and so it's a fairly intuitive thing because you basically need sufficiently many degrees of freedom in the controller uh to be able to map to to model what's in the environment um so you can see this as this degeneracy um property of resilient systems or uh any system under the FP as an instance of this law of of requisite variety it's also clearly by the way a an obvious case of constrained maximum entropy so there's work um showing that many many lines of thought really showing that the active inference uh sorry the free energy principle is um is equivalent to a principle of constrained maximum entropy uh so here the constraint is this uh you can you can think of this as constrained by accuracy um but you you basically maximize this entropy term subject to that constraint uh subject to the generative model okay so um looking just a bit more closely at this notion of degeneracy um so I've said that it's Central um one pointed question you might ask here is well are all do all resilient systems uh have this degeneracy property uh and so I want to say the answer is yes um so the type of degeneracy that we've already talked about is this degeneracy and the likelihood mapping uh so this is multiple States mapping to the same observation um and again that's uh that can be cast as you'd expect to see that given um that you're performing maximum entropy inference um so that that is um crucial but equally important I'd say is degeneracy and the Dynamics uh so this characterizes just systems that again that have an attracting set of States so if these STS are states at a given time and then $st+$ one is the state or set of States at the next time step um so this in order for for a system to be resilient um it really has to be the case that this transition um uh that these transition probabilities or the Dynamics of the system are such that um that that mapping itself is degenerate um so you know this means that basically you can perturb the system in several different ways and you'll still arrive uh you'll still end up at the same at the same uh state or or set of attracting States um obviously if the if the perturbation is too large then that might not be the case the system will diverge and you'll destroy you'll destroy it but by definition the systems that survive through change are the ones that have this property um so there is a very deep link between resilience and genery if you consider Dynamics as well um just a couple more points here one is um something that I've recently been thinking about uh it's it's it's actually a theme that's been on my mind for some time but I've recently written about it in some other work um and this is what what does the principle of requisite variety and um what

do these concepts of degeneracy and resilience look like in in the context of multiscale active inference um but specifically here I don't I don't just mean um the fact that excuse me the fact that model parameters as well as states can change but I mean U multiscale active inference in the sense that you've got potentially higher level agents that are composed of lower level agents um so in that kind of setting uh requisite variety ends up looking like the the property of higher level agents that they're composed of a variety of different types of lower level agents um so I mean I would argue that if you combine this law of requisite variety with multiscale active inference in this sense then you really sort of automatically have something like a a diverse ecosystem in which there are many different types of agents that participate um so here in this diagram we're illustrating first of all just the same degenerate State observation mapping that we saw in the uh in the previous slides um but here the nodes are individual agents that compose this higher level agent um also as pointed out in this paper um by by um Carl friston and Daniel fredman at all um the in this kind of context different agents can be expected to and I think empirically in some simulations are shown to take on the roles of these different types of States so uh of course the breakdown here is in terms of active and sensory and uh active sensory and internal States so here you've got active sensory and internal agents uh and that's one I would argue of this um application of the of the law of requisite variety um to this higher level uh system um I also show here the the policy selection based on expected free energy for completeness um but that's sort of the main the main point is this this specialization and diversification of Agents um in this in this setting um in terms of the second definition of degeneracy you can also think of these agents as playing um somewhat distinct but partially overlapping roles um and there's some some lots of interesting work on um on that sort of multi-agent perspective from the point of view of active inference um so that's really it I'm just going to conclude with some with sort of a philosophical question that I think is is kind of important for um for tying this this together and and sort of checking that it that it just makes sense um so I've said earlier that um any system well described by the FP as res so hold on a second uh not all systems are equally resilient though uh right tups by are sort of the paradigm of a fragile as opposed to resilient object so since the FP describes things in general doesn't it mean that all things are resilient um and doesn't that contradict this obvious fact so I think the answer here is that it's simply it's a matter of scale so yes all things are resilient all things that is things that are reidentified over time and that persist through change um all such things are resilient but it's a matter of scale so if you if you introduce a very slight perturbation to this teacup um very slight you know um relative to a certain a

certain reference frame or a certain scale but large enough right relative to the scale on which that system is resilient then by definition it will survive right and so obviously again in a in a a real universe in which there are interactions and you're you're always interacting in some way dealing with some sort of um random fluctuations um then yes for the system to persist through time to be inert and elastic um it has some right the material has some uh coherence that allows it to to persist through those changes so it is a resilient system it's just that if you introduce a change that's um on a scale that's That's too great for the system to handle then it'll be destroyed um but sort of the just the basic coherence as a spao temporal object of this teacup is itself a form of resilience uh and then we get when we get into more complex forms of life more complex generative models that's when we when we when we begin to see more um impressive forms of resilience and shading it even into into plasticity and the ability to to really survive through more significant changes so thanks for for listening SL watching um I'm happy to participate in this Symposium um thanks to all the other uh the co-authors of this um paper by Mark Mall in 2022 that was the departure point for this work uh yeah thank you